

From First Principles to Modern Large Language Models

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Contents

Block A — Foundations	2
Chapter 1: Sets, functions, logic, proofs	2
Chapter 2: Numbers, sequences, limits, completeness	4
Chapter 3: Continuity, univariate differentiation, chain rule	6
Chapter 4: Multivariate calculus: partials, gradients, Jacobians	9
Chapter 5: Linear algebra I: vector spaces, basis, linear maps	11
Chapter 6: Linear algebra II: inner products, norms, eigenvalues, SVD	13
Chapter 7: Convexity and optimization; gradient descent convergence	16
Block B — Probability and Information	18
Chapter 8: Probability foundations: sample spaces, sigma-algebras, Kolmogorov axioms	18
Chapter 9: Random variables, distributions, CDF/PMF/PDF	18
Chapter 10: Expectation, variance, covariance; Jensen's inequality	18
Chapter 11: Information theory: self-information, entropy, cross-entropy, KL	18
Chapter 12: Statistical inference: likelihood, MLE, ERM, bias-variance	19
Block C — Stochastic Optimization	19
Chapter 13: SGD: stochastic-approximation theorem; mini-batching; convergence sketch	19
Chapter 14: Momentum, RMSProp, AdamW: derivation and bias-correction proof	19
Block D — Neural Networks	19
Chapter 15: MLPs as compositional functions; universal approximation	19
Chapter 16: Activation functions: ReLU/GELU/softmax with derivatives	19

Chapter 17: Loss functions: MSE, cross-entropy; gradients from first principles	19
Chapter 18: Backpropagation: chain rule applied; reverse-mode AD as a graph algorithm	19
Block E — Sequence Models and Attention	20
Chapter 19: Embeddings: token to vector; lookup as a linear map; weight tying	20
Chapter 20: RNN intuition; vanishing-gradient proof; why we need attention	20
Chapter 21: Scaled dot-product attention: derivation, softmax-temperature analysis	20
Chapter 22: Multi-head attention: parallel heads as concat-then-project; complexity	20
Chapter 23: Transformer block: residual + LayerNorm/RMSNorm + FFN + attention; gradient-flow argument	20
Chapter 24: Positional encoding: sinusoidal derivation, RoPE construction	20
Block F — Pre-training	20
Chapter 25: Causal masking; next-token prediction loss as MLE on the empirical distribution	20
Chapter 26: Tokenization: BPE algorithm; greedy merge correctness	20
Chapter 27: Pre-training pipeline: AdamW + warmup + cosine decay + gradient clipping; tiny-GPT training run	21
Block G — Post-training	21
Chapter 28: SFT, RLHF (PPO/GRPO), and DPO; train + post-train a tiny GPT	21

Block A — Foundations

Chapter 1: Sets, functions, logic, proofs

Motivation

Every object in this book—a vector, a probability distribution, a neural network, a token—is ultimately a *set with structure*. Before we can build the embedding map of Chapter 19 or even define the real numbers in Chapter 5, we need a precise grammar for membership, functions, and proof. Chapter 8 (linear maps) treats functions between vector spaces; Chapter 15 (probability) treats measures on σ -algebras of sets. Both collapse without the vocabulary below. We therefore start, with no apology, from the bottom.

Definitions

Definition 0.1 (Set, membership, subset). A set S is a collection of distinct objects called elements. We write $x \in S$ when x is an element of S , and $x \notin S$ otherwise. The empty set $\emptyset$ is the unique set with no elements. We say A is a subset of B , written $A \subset B$, iff $\forall x (x \in A \Rightarrow x \in B)$. Two sets are equal, $A = B$, iff $A \subset B$ and $B \subset A$.

Definition 0.2 (Boolean operations). Given A, B inside a universe U :

$$\begin{aligned} A \cup B &:= \{x \in U : x \in A \vee x \in B\}, \\ A \cap B &:= \{x \in U : x \in A \wedge x \in B\}, \\ A^c &:= \{x \in U : x \notin A\}, \\ \mathcal{P}(S) &:= \{T : T \subset S\}. \end{aligned}$$

Definition 0.3 (Function, image, preimage). A function $f : A \rightarrow B$ assigns to each $a \in A$ exactly one $f(a) \in B$. The image is $f(A) := \{f(a) : a \in A\}$ and the preimage of $T \subset B$ is $f^{-1}(T) := \{a \in A : f(a) \in T\}$. We call f :

- injective iff $\forall a_1, a_2 \in A, f(a_1) = f(a_2) \Rightarrow a_1 = a_2$;
- surjective iff $\forall b \in B, \exists a \in A, f(a) = b$;
- bijective iff both.

Remark 0.4 (Logic). Propositional connectives: $\wedge, \vee, \neg, \Rightarrow, \Leftrightarrow$. Quantifiers: $\forall, \exists$. Standard proof techniques: *direct* (assume P , derive Q), *contrapositive* ($P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$), *contradiction* (assume $\neg P$, derive $\perp$), and *induction* (Theorem 0.7).

Theorems and proofs

Theorem 0.5 (De Morgan's laws for sets). For sets $A, B \subset U$,

$$(A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (A \cap B)^c = A^c \cup B^c.$$

Proof. We prove the first; the second is symmetric. Fix arbitrary $x \in U$. We show the biconditional $x \in (A \cup B)^c \Leftrightarrow x \in A^c \cap B^c$.

($\Rightarrow$) Assume $x \in (A \cup B)^c$. By definition, $x \notin A \cup B$, i.e. $\neg(x \in A \vee x \in B)$. By the propositional De Morgan law $\neg(P \vee Q) \equiv \neg P \wedge \neg Q$, this is $\neg(x \in A) \wedge \neg(x \in B)$, i.e. $x \notin A$ and $x \notin B$. Hence $x \in A^c$ and $x \in B^c$, so $x \in A^c \cap B^c$.

($\Leftarrow$) Assume $x \in A^c \cap B^c$. Then $x \notin A$ and $x \notin B$, so $\neg(x \in A) \wedge \neg(x \in B)$, which by propositional De Morgan equals $\neg(x \in A \vee x \in B)$, i.e. $x \notin A \cup B$, i.e. $x \in (A \cup B)^c$.

For the second identity, the same argument with $\wedge$ and $\vee$ interchanged (using $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$) gives $x \in (A \cap B)^c \Leftrightarrow x \in A^c \cup B^c$. $\square$

Theorem 0.6 (Composition of injections is injective). If $f : A \rightarrow B$ and $g : B \rightarrow C$ are injective, then $g \circ f : A \rightarrow C$ is injective.

Proof. Let $a_1, a_2 \in A$ with $(g \circ f)(a_1) = (g \circ f)(a_2)$, i.e. $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. By definition $g \circ f$ is injective. $\square$

Theorem 0.7 (Principle of mathematical induction). Let $P(n)$ be a predicate on $\mathbb{N} = \{1, 2, 3, \dots\}$. If

- (i) $P(1)$ holds, and
- (ii) $\forall n \in \mathbb{N}, P(n) \Rightarrow P(n+1)$,

then $\forall n \in \mathbb{N}, P(n)$.

Proof. We assume the well-ordering principle: every nonempty subset of $\mathbb{N}$ has a least element. Let

$$S := \{n \in \mathbb{N} : \neg P(n)\}.$$

We show $S = \emptyset$ by contradiction. Suppose $S \neq \emptyset$. By well-ordering, S has a least element m . By (i), $P(1)$ holds, so $1 \notin S$, hence $m \geq 2$, so $m-1 \in \mathbb{N}$. By minimality of m in S , $m-1 \notin S$, i.e. $P(m-1)$ holds. By (ii) applied at $n = m-1$, $P(m)$ holds, so $m \notin S$. This contradicts $m \in S$. Therefore $S = \emptyset$, i.e. $P(n)$ holds for every $n \in \mathbb{N}$. $\square$

Code sketch

In the companion notebook we (a) enumerate $\mathcal{P}(\{a, b, c\})$ from scratch and verify $|\mathcal{P}(S)| = 2^{|S|}$; (b) brute-force De Morgan on $U = \{1, \dots, 8\}$ with $A = \{1, 3, 5, 7\}$, $B = \{2, 3, 5, 7\}$; (c) implement `is_injective`, `is_surjective`, `is_bijective` as predicates over finite functions encoded as Python dictionaries; (d) verify $\sum_{k=1}^n k = n(n+1)/2$ both by direct summation up to $n = 20$ and by checking the induction step $S(n+1) - S(n) = n+1$ numerically.

Connection to LLMs

A language model's *vocabulary* $\mathcal{V}$ is a finite set of tokens (typically $|\mathcal{V}| \approx 5 \times 10^4$). The tokenizer is a function $T : \text{strings} \rightarrow \mathcal{V}^*$. The *embedding map* $E : \mathcal{V} \rightarrow \mathbb{R}^d$ is a function from a finite set into a real vector space; the implementation as a lookup table $E[i]$ requires that the vocabulary index $\mathcal{V} \rightarrow \{0, 1, \dots, |\mathcal{V}| - 1\}$ be a *bijection*. Injectivity guarantees distinct tokens get distinct rows; surjectivity guarantees every row is reachable. We revisit this in Chapter 19, where E becomes the first learnable layer of the transformer. The rest of the book is, in a strict sense, a long story about which functions between which sets one is allowed to write down.

Chapter 2: Numbers, sequences, limits, completeness

Motivation

In Chapter 1 we built logic, sets, and proof technique. We now build the arithmetic playground in which all of analysis, optimization, and ultimately neural network training takes place: the real numbers $\mathbb{R}$. The central question of this chapter is not “what are numbers?” in a metaphysical sense but rather: *what structural property of $\mathbb{R}$ makes it the right home for limits?* The answer is **completeness**, and it is what will let us, many chapters later, prove that gradient descent iterates $\theta_t \in \mathbb{R}^d$ actually converge.

Definitions

We accept $\mathbb{N} = \{0, 1, 2, \dots\}$ as constructed in Chapter 1 (von Neumann ordinals). The integers $\mathbb{Z}$ extend $\mathbb{N}$ by formal additive inverses; the rationals $\mathbb{Q}$ extend $\mathbb{Z}$ by formal multiplicative inverses of nonzero elements. We then take $\mathbb{R}$ to be a complete ordered field containing $\mathbb{Q}$ (constructed via Dedekind cuts or Cauchy completion of $\mathbb{Q}$). Thus

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Definition 0.8 (Sequence). A sequence in a set X is a function $a : \mathbb{N} \rightarrow X$, written $(a_n)_{n \in \mathbb{N}}$.

Definition 0.9 (Limit, ε - N). A real sequence (a_n) converges to $L \in \mathbb{R}$, written $a_n \rightarrow L$, iff

$$\forall \varepsilon > 0 \quad \exists N \in \mathbb{N} \quad \forall n \geq N : |a_n - L| < \varepsilon.$$

Definition 0.10 (Cauchy). (a_n) is Cauchy iff $\forall \varepsilon > 0 \quad \exists N \quad \forall m, n \geq N : |a_m - a_n| < \varepsilon$.

Definition 0.11 (Bounded, monotone). (a_n) is bounded above iff $\exists M : a_n \leq M$ for all n ; bounded below symmetrically; bounded iff both. It is monotone increasing iff $a_n \leq a_{n+1}$ for all n .

Definition 0.12 (Supremum, infimum). Let $S \subseteq \mathbb{R}$ be nonempty and bounded above. $u \in \mathbb{R}$ is an upper bound if $s \leq u$ for all $s \in S$. The supremum $\sup S$ is the least upper bound: an upper bound such that for every $\varepsilon > 0$ there exists $s \in S$ with $s > \sup S - \varepsilon$. The infimum $\inf S$ is defined dually.

Definition 0.13 (Completeness axiom of $\mathbb{R}$). Every nonempty subset of $\mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Theorems and proofs

Theorem 0.14 (Irrationality of $\sqrt{2}$). There is no rational q with $q^2 = 2$.

Proof. Suppose for contradiction (technique from Chapter 1) that $q = p/r$ with $p, r \in \mathbb{Z}$, $r \neq 0$, the fraction in lowest terms (so $\gcd(p, r) = 1$), and $q^2 = 2$. Then $p^2 = 2r^2$, so p^2 is even, hence p is even (since the square of an odd integer is odd). Write $p = 2k$. Substituting, $4k^2 = 2r^2$, i.e. $r^2 = 2k^2$, so r^2 is even and therefore r is even. But then $2 \mid \gcd(p, r)$, contradicting $\gcd(p, r) = 1$. $\square$

Remark 0.15. This is the first concrete witness that $\mathbb{Q}$ is *incomplete*: the bounded-above set $S = \{q \in \mathbb{Q} : q^2 < 2\}$ has no supremum in $\mathbb{Q}$. The completeness axiom is exactly the patch that distinguishes $\mathbb{R}$ from $\mathbb{Q}$.

Theorem 0.16 (Bounded monotone convergence). Let (a_n) be monotone increasing and bounded above in $\mathbb{R}$. Then (a_n) converges, and $\lim a_n = \sup_n a_n$.

Proof. Let $S = \{a_n : n \in \mathbb{N}\}$. S is nonempty (it contains a_0) and bounded above by hypothesis. By the completeness axiom, $L := \sup S$ exists in $\mathbb{R}$. Fix $\varepsilon > 0$. By the supremum's least-upper-bound property, $L - \varepsilon$ is *not* an upper bound, so there exists $N \in \mathbb{N}$ with $a_N > L - \varepsilon$. For any $n \geq N$, monotonicity gives $a_n \geq a_N > L - \varepsilon$, while $a_n \leq L$ since L is an upper bound. Hence

$$L - \varepsilon < a_n \leq L \quad \implies \quad |a_n - L| < \varepsilon,$$

which is exactly the ε - N definition of $\lim a_n = L$. $\square$

Lemma 0.17 (Bolzano–Weierstrass for $\mathbb{R}$). Every bounded real sequence has a convergent subsequence.

Proof. Let $(a_n) \subseteq [c_0, d_0]$ with $c_0 < d_0$. Bisect at midpoint $m_0 = (c_0 + d_0)/2$; at least one of $[c_0, m_0]$, $[m_0, d_0]$ contains a_n for infinitely many n (else only finitely many terms total, contradicting $|\mathbb{N}| = \infty$). Call this half $[c_1, d_1]$ and pick an index n_1 with $a_{n_1} \in [c_1, d_1]$. Iterate: at stage k we have nested $[c_k, d_k]$ of length $(d_0 - c_0)/2^k$ each containing infinitely many terms, and pick $n_k > n_{k-1}$ with $a_{n_k} \in [c_k, d_k]$. The sequence (c_k) is monotone increasing and bounded above by d_0 , so by Theorem 2.8 it converges to some L ; similarly $(d_k) \rightarrow L$ since $d_k - c_k = (d_0 - c_0)/2^k \rightarrow 0$. Since $c_k \leq a_{n_k} \leq d_k$, the squeeze gives $a_{n_k} \rightarrow L$. $\square$

Theorem 0.18 (Cauchy completeness of $\mathbb{R}$). *Every Cauchy sequence in $\mathbb{R}$ converges.*

Proof. Let (a_n) be Cauchy.

Step 1 (bounded). Pick $\varepsilon = 1$ and N from the Cauchy condition; for $n \geq N$, $|a_n| \leq |a_N| + 1$, so (a_n) is bounded by $M = \max(|a_0|, \dots, |a_{N-1}|, |a_N| + 1)$.

Step 2 (convergent subsequence). By the Bolzano–Weierstrass lemma, some subsequence $a_{n_k} \rightarrow L$.

Step 3 (full sequence). Fix $\varepsilon > 0$. Choose N_1 with $|a_m - a_n| < \varepsilon/2$ for all $m, n \geq N_1$ (Cauchy property), and choose K with $n_K \geq N_1$ and $|a_{n_K} - L| < \varepsilon/2$ (subsequence convergence). Then for any $n \geq N_1$,

$$|a_n - L| \leq |a_n - a_{n_K}| + |a_{n_K} - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon. \quad \square$$

Code sketch

Numerically the partial sums $S_n = \sum_{k=1}^n 1/k^2$ approach $\pi^2/6$. We print $|S_n - \pi^2/6|$ and locate the smallest N realizing each tolerance $\varepsilon \in \{10^{-2}, 10^{-4}, 10^{-6}\}$, contrast a Cauchy sequence $(1/n)$ with a non-Cauchy one $((-1)^n)$, run a Newton iteration to $\sqrt{2}$, and watch a bounded monotone telescoping series converge to 1.

Connection to LLMs

Training a transformer is a sequence $(\theta_t)_{t \in \mathbb{N}}$ in $\mathbb{R}^d$ produced by an optimizer (SGD, Adam). Statements like “the loss converges” or “the iterates converge to a stationary point” are *limit statements in the sense of Definition 2.2*. Convergence proofs for SGD (Chapter 13) and Adam (Chapter 14) reduce, ultimately, to bounded-monotone arguments on auxiliary scalar sequences (loss values, squared gradient norms) — and those arguments would simply be false over $\mathbb{Q}$. Completeness is not philosophical decoration; it is the load-bearing axiom under every convergence theorem in deep learning.

Chapter 3: Continuity, univariate differentiation, chain rule

Motivation

Chapter 2 built the language of limits (ε – N for sequences and ε – δ for functions). *Continuity* asserts that small input perturbations cause small output perturbations; *differentiability* strengthens this to “the perturbation is asymptotically linear.” Both notions sit beneath every gradient that flows through a neural network. The chain rule, the central theorem of this chapter, is precisely the algebraic identity that backpropagation evaluates at scale (forward to Chapter 18).

Definitions

Definition 0.19 (ε – δ continuity at a point). *Let $f : D \subseteq \mathbb{R} \rightarrow \mathbb{R}$ and $a \in D$. We say f is continuous at a iff*

$$\forall \varepsilon > 0 \exists \delta > 0 : |x - a| < \delta \text{ and } x \in D \implies |f(x) - f(a)| < \varepsilon.$$

f is continuous on $S \subseteq D$ iff it is continuous at every $a \in S$.

Proposition 0.20 (Sequential characterization). *f is continuous at a iff for every sequence $x_n \rightarrow a$ in D we have $f(x_n) \rightarrow f(a)$.*

Proof. ($\Rightarrow$) Given $\varepsilon > 0$, pick δ from continuity; since $x_n \rightarrow a$, eventually $|x_n - a| < \delta$, hence $|f(x_n) - f(a)| < \varepsilon$. ($\Leftarrow$) Suppose f is not continuous at a : some $\varepsilon_0 > 0$ admits no working δ , so for each $n \in \mathbb{N}$ pick x_n with $|x_n - a| < 1/n$ but $|f(x_n) - f(a)| \geq \varepsilon_0$. Then $x_n \rightarrow a$ and yet $f(x_n) \not\rightarrow f(a)$, contradicting the hypothesis. $\square$

Definition 0.21 (Differentiability). *For a an interior point of D , f is differentiable at a iff*

$$f'(a) := \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

exists in $\mathbb{R}$.

Theorems and proofs

Theorem 0.22 (Differentiable $\Rightarrow$ continuous). *If f is differentiable at a , then f is continuous at a .*

Proof. For $h \neq 0$ we have the algebraic identity

$$f(a+h) - f(a) = h \cdot \frac{f(a+h) - f(a)}{h}.$$

By limit algebra (Chapter 2), the right-hand side tends to $0 \cdot f'(a) = 0$ as $h \rightarrow 0$. Hence $\lim_{h \rightarrow 0} f(a+h) = f(a)$, which is continuity at a . $\square$

Theorem 0.23 (Sum, product, quotient rules). *Let f, g be differentiable at a . Then $f + g$, fg , and (provided $g(a) \neq 0$) f/g are differentiable at a , with*

$$(f+g)'(a) = f'(a) + g'(a), \quad (fg)'(a) = f'(a)g(a) + f(a)g'(a),$$

$$\left(\frac{f}{g}\right)'(a) = \frac{f'(a)g(a) - f(a)g'(a)}{g(a)^2}.$$

Proof of the product rule. Use the add-and-subtract trick:

$$\begin{aligned} \frac{(fg)(a+h) - (fg)(a)}{h} &= \frac{f(a+h)g(a+h) - f(a)g(a+h)}{h} + \frac{f(a)g(a+h) - f(a)g(a)}{h} \\ &= \frac{f(a+h) - f(a)}{h} g(a+h) + f(a) \frac{g(a+h) - g(a)}{h}. \end{aligned}$$

By Theorem 0.22, $g(a+h) \rightarrow g(a)$ as $h \rightarrow 0$. The two difference quotients tend to $f'(a)$ and $g'(a)$ respectively. Limit algebra yields $f'(a)g(a) + f(a)g'(a)$. Sum and quotient rules are analogous (use the algebraic identity $\frac{1}{g(a+h)} - \frac{1}{g(a)} = -\frac{g(a+h)-g(a)}{g(a+h)g(a)}$ and continuity of g). $\square$

Theorem 0.24 (Chain rule, Carathéodory form). *Let g be differentiable at a and f differentiable at $b := g(a)$. Then $f \circ g$ is differentiable at a and*

$$(f \circ g)'(a) = f'(g(a)) \cdot g'(a).$$

Proof. Define the Carathéodory function

$$\phi(y) := \begin{cases} \frac{f(y) - f(b)}{y - b}, & y \neq b, \\ f'(b), & y = b. \end{cases}$$

By the definition of $f'(b)$, ϕ is continuous at b . For *every* y in a neighborhood of b ,

$$f(y) - f(b) = \phi(y)(y - b), \quad (\star)$$

which holds at $y = b$ trivially and by construction otherwise — a key advantage: no division by zero is ever required. Apply $(\star)$ with $y = g(x)$:

$$\frac{f(g(x)) - f(g(a))}{x - a} = \phi(g(x)) \cdot \frac{g(x) - g(a)}{x - a}.$$

As $x \rightarrow a$, $g(x) \rightarrow g(a) = b$ by Theorem 0.22, and ϕ is continuous at b , so $\phi(g(x)) \rightarrow \phi(b) = f'(b)$ by Proposition 0.20. The second factor tends to $g'(a)$. Limit algebra closes the proof. $\square$

Remark 0.25. The Carathéodory device sidesteps the classical pitfall of the difference-quotient proof, where one must handle the case $g(x) = g(a)$ for x arbitrarily close to a (e.g. g constant on a sequence). Here ϕ is defined everywhere, so the identity $(\star)$ requires no caveat.

Lemma 0.26 (Rolle). *If f is continuous on $[a, b]$, differentiable on (a, b) , and $f(a) = f(b)$, then there exists $c \in (a, b)$ with $f'(c) = 0$.*

Proof. By the extreme value theorem (Chapter 4, via compactness of $[a, b]$), f attains its maximum M and minimum m on $[a, b]$. If $M = m$ then f is constant and any $c \in (a, b)$ works. Otherwise an extremum is attained at some interior $c \in (a, b)$; Fermat's interior-extremum lemma — one-sided difference quotients have opposite signs at an interior extremum, so the two-sided limit $f'(c)$ must equal 0 — finishes the job. $\square$

Theorem 0.27 (Mean value theorem). *If f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists $c \in (a, b)$ with*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Proof. Define the auxiliary function

$$h(x) := f(x) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then h is continuous on $[a, b]$ (sum/product of continuous functions) and differentiable on (a, b) with

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Direct computation gives $h(a) = f(a)$ and $h(b) = f(b) - (f(b) - f(a)) = f(a)$, so $h(a) = h(b)$. By Lemma 0.26 there exists $c \in (a, b)$ with $h'(c) = 0$, i.e. $f'(c) = (f(b) - f(a))/(b - a)$. $\square$

Code sketch

The companion notebook (`cells.json`) (i) certifies ε - δ continuity for $f(x) = x^2$ at $a = 2$ by binary-search on δ for $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$; (ii) plots the $O(h)$ error of the forward-difference approximation to $\sin'$; (iii) verifies the chain rule for $f(x) = \sin(x^2)$ at $x = 1$ to $\sim 10^{-7}$; (iv) finds an MVT witness $c \in (0, 2)$ for $f(x) = x^3 - 2x$ by bisection.

Connection to LLMs

A transformer is a composition $L_K \circ L_{K-1} \circ \cdots \circ L_1$ of differentiable layers. The gradient of the loss with respect to any parameter θ in layer i is, by iterated application of Theorem 0.24,

$$\frac{\partial \mathcal{L}}{\partial \theta} = \frac{\partial \mathcal{L}}{\partial L_K} \cdot \frac{\partial L_K}{\partial L_{K-1}} \cdots \frac{\partial L_{i+1}}{\partial L_i} \cdot \frac{\partial L_i}{\partial \theta}.$$

Backpropagation is precisely the right-to-left evaluation of this product (Chapter 18). The MVT, in turn, underwrites convergence proofs for gradient descent (Chapter 14): it bounds $|f(x) - f(y)| \leq \sup |f'| \cdot |x - y|$, the Lipschitz hypothesis under which descent provably decreases the loss. Without the chain rule, deep learning does not exist.

Chapter 4: Multivariate calculus: partials, gradients, Jacobians

Motivation

A neural network is a function $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ built by composing many simple maps; training it requires the gradient of a scalar loss with respect to a high-dimensional parameter vector. Chapter 3 handled the one-dimensional theory: existence of $f'(a)$, the mean value theorem (MVT), and the single-variable chain rule through the Carathéodory factorization $f(x) - f(a) = \varphi(x)(x - a)$ with φ continuous at a . We now lift these ideas to several variables. Two pitfalls: (i) the *correct* notion of differentiability is *not* “all partials exist” but the stronger Fréchet condition, and (ii) the chain rule becomes a matrix product—the operation iterated by backpropagation in Chapter 18.

Definitions

Let $f : U \rightarrow \mathbb{R}^m$ with $U \subseteq \mathbb{R}^n$ open, $\mathbf{a} \in U$, and $\mathbf{e}_i$ the i -th standard basis vector.

Definition 0.28 (Partial derivative). $\frac{\partial f}{\partial x_i}(\mathbf{a}) := \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a})}{h}$, when the limit exists.

Definition 0.29 (Directional derivative). For a unit vector $\mathbf{v}$, $D_{\mathbf{v}}f(\mathbf{a}) := \lim_{t \rightarrow 0} \frac{f(\mathbf{a} + t\mathbf{v}) - f(\mathbf{a})}{t}$.

Definition 0.30 (Fréchet differentiability). f is differentiable at $\mathbf{a}$ if there exists a linear $L : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{\|f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) - L\mathbf{h}\|}{\|\mathbf{h}\|} = 0,$$

i.e. the remainder is $o(\|\mathbf{h}\|)$. The matrix of L in the standard basis is the Jacobian $J_f(\mathbf{a}) \in \mathbb{R}^{m \times n}$. When $m = 1$, the gradient is $\nabla f(\mathbf{a}) := J_f(\mathbf{a})^\top$.

Remark 0.31. Differentiability is strictly stronger than existence of partials: $f(x, y) = xy/(x^2 + y^2)$ (with $f(0, 0) = 0$) has both partials zero at the origin, yet f is discontinuous there and so not Fréchet differentiable.

Theorems and proofs

Theorem 0.32 (Differentiable implies partials exist; $L = J_f$). If f is differentiable at $\mathbf{a}$ with derivative L , then $\partial f / \partial x_j(\mathbf{a})$ exists for every j and equals the j -th column of the matrix of L .

Proof. Take $\mathbf{h} = h\mathbf{e}_j$, $h \neq 0$. Differentiability yields

$$f(\mathbf{a} + h\mathbf{e}_j) - f(\mathbf{a}) = hL\mathbf{e}_j + r(h), \quad \|r(h)\|/|h| \rightarrow 0.$$

Divide by h and let $h \rightarrow 0$: the left side tends to $\partial f/\partial x_j(\mathbf{a})$ by Definition 1, the right side to $L\mathbf{e}_j$. Hence the partial exists and equals the j -th column of L . $\square$

Theorem 0.33 (Continuous partials imply differentiability). *If all $\partial f/\partial x_j$ exist on a neighborhood of $\mathbf{a}$ and are continuous at $\mathbf{a}$, then f is differentiable at $\mathbf{a}$.*

Sketch. Reduce to scalar codomain componentwise. Telescope along axes:

$$f(\mathbf{a} + \mathbf{h}) - f(\mathbf{a}) = \sum_{j=1}^n \left[f(\mathbf{a} + \mathbf{h}_{<j} + h_j\mathbf{e}_j) - f(\mathbf{a} + \mathbf{h}_{<j}) \right], \quad \mathbf{h}_{<j} := \sum_{i<j} h_i\mathbf{e}_i.$$

The single-variable MVT (Chapter 3) applied to the j -th summand produces ξ_j between 0 and h_j with

$$f(\mathbf{a} + \mathbf{h}_{<j} + h_j\mathbf{e}_j) - f(\mathbf{a} + \mathbf{h}_{<j}) = h_j \partial_j f(\mathbf{a} + \mathbf{h}_{<j} + \xi_j\mathbf{e}_j).$$

Subtract $\sum_j h_j \partial_j f(\mathbf{a})$. By continuity of each $\partial_j f$, the difference $|\partial_j f(\cdot) - \partial_j f(\mathbf{a})| \rightarrow 0$ as $\mathbf{h} \rightarrow \mathbf{0}$. Cauchy–Schwarz then bounds the remainder by $\|\mathbf{h}\| \cdot \varepsilon(\mathbf{h})$ with $\varepsilon \rightarrow 0$, i.e. $o(\|\mathbf{h}\|)$. (Spivak, *Calculus on Manifolds*, Thm. 2-8.) $\square$

Theorem 0.34 (Multivariate chain rule). *If $g : \mathbb{R}^k \rightarrow \mathbb{R}^n$ is differentiable at $\mathbf{a}$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is differentiable at $\mathbf{b} = g(\mathbf{a})$, then $f \circ g$ is differentiable at $\mathbf{a}$ with*

$$J_{f \circ g}(\mathbf{a}) = J_f(\mathbf{b}) J_g(\mathbf{a}).$$

Proof. Write the Carathéodory-style remainders $\rho_g(\mathbf{h}) := g(\mathbf{a} + \mathbf{h}) - g(\mathbf{a}) - J_g(\mathbf{a})\mathbf{h} = o(\|\mathbf{h}\|)$, and $\rho_f(\mathbf{k}) = o(\|\mathbf{k}\|)$. Let $\mathbf{k}(\mathbf{h}) := g(\mathbf{a} + \mathbf{h}) - g(\mathbf{a})$. Then

$$\begin{aligned} f(g(\mathbf{a} + \mathbf{h})) - f(g(\mathbf{a})) &= J_f(\mathbf{b})\mathbf{k}(\mathbf{h}) + \rho_f(\mathbf{k}(\mathbf{h})) \\ &= J_f(\mathbf{b})J_g(\mathbf{a})\mathbf{h} + J_f(\mathbf{b})\rho_g(\mathbf{h}) + \rho_f(\mathbf{k}(\mathbf{h})). \end{aligned}$$

Bound: $\|J_f(\mathbf{b})\rho_g(\mathbf{h})\| \leq \|J_f(\mathbf{b})\|_{\text{op}} \cdot o(\|\mathbf{h}\|) = o(\|\mathbf{h}\|)$. For the last term, $\|\mathbf{k}(\mathbf{h})\| \leq (\|J_g(\mathbf{a})\|_{\text{op}} + 1)\|\mathbf{h}\|$ for small $\|\mathbf{h}\|$, hence $\rho_f(\mathbf{k}(\mathbf{h})) = o(\|\mathbf{k}(\mathbf{h})\|) = o(\|\mathbf{h}\|)$. The total remainder is $o(\|\mathbf{h}\|)$, so $f \circ g$ is differentiable with derivative $J_f(\mathbf{b})J_g(\mathbf{a})$. $\square$

Theorem 0.35 (Schwarz / Clairaut). *If $f : U \rightarrow \mathbb{R}$ is C^2 on U open and $\mathbf{a} \in U$, then for all i, j*

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) = \frac{\partial^2 f}{\partial x_j \partial x_i}(\mathbf{a}).$$

Proof. Fix $i \neq j$ and restrict to the two-variable slice in coordinates x_i, x_j (other coordinates frozen at the values of $\mathbf{a}$). Define the second difference

$$\Delta(h, k) := f(\mathbf{a} + h\mathbf{e}_i + k\mathbf{e}_j) - f(\mathbf{a} + h\mathbf{e}_i) - f(\mathbf{a} + k\mathbf{e}_j) + f(\mathbf{a}).$$

Let $\varphi(s) := f(\mathbf{a} + s\mathbf{e}_i + k\mathbf{e}_j) - f(\mathbf{a} + s\mathbf{e}_i)$, so $\Delta(h, k) = \varphi(h) - \varphi(0)$. By MVT in s : $\Delta = h\varphi'(\xi)$ for some $\xi \in (0, h)$, with

$$\varphi'(\xi) = \partial_i f(\mathbf{a} + \xi\mathbf{e}_i + k\mathbf{e}_j) - \partial_i f(\mathbf{a} + \xi\mathbf{e}_i).$$

Apply MVT again, this time in k : $\varphi'(\xi) = k\partial_j \partial_i f(\mathbf{a} + \xi\mathbf{e}_i + \eta\mathbf{e}_j)$ for some $\eta \in (0, k)$. Hence

$$\Delta(h, k) = hk \partial_j \partial_i f(\mathbf{a} + \xi\mathbf{e}_i + \eta\mathbf{e}_j).$$

By the symmetric argument starting with $\psi(t) := f(\mathbf{a} + h\mathbf{e}_i + t\mathbf{e}_j) - f(\mathbf{a} + t\mathbf{e}_j)$, $\Delta(h, k) = hk \partial_i \partial_j f(\mathbf{a} + \xi'\mathbf{e}_i + \eta'\mathbf{e}_j)$. Divide by hk and let $(h, k) \rightarrow (0, 0)$. Continuity of the second partials (C^2) makes both sides converge to $\partial_j \partial_i f(\mathbf{a})$ and $\partial_i \partial_j f(\mathbf{a})$ respectively, which are therefore equal. $\square$

Code sketch

The accompanying notebook verifies every theorem numerically: gradient of $f(x, y) = x^2y + \sin(x + y)$ via central differences; Jacobian of $\mathbf{f}(x, y, z) = (xyz, x^2 + \sin y + e^z)$ column-by-column; chain rule on $f \circ g$ with $g(t) = (\cos t, \sin t)$, $f(x, y) = x^2 + y^2$ (composite identically 1, so the chain-rule product must vanish); equality of mixed partials for $f(x, y) = x^3y^2 + \sin(xy)$.

Connection to LLMs

A transformer is a composition $F = F_L \circ \cdots \circ F_1$ of differentiable layers, scored by a scalar loss $\mathcal{L} = \ell \circ F$. Theorem 4.7 states

$$\nabla_{\theta_\ell} \mathcal{L} = (J_\ell(F(x)) J_{F_L} \cdots J_{F_{\ell+1}}) \frac{\partial F_\ell}{\partial \theta_\ell}.$$

Backpropagation never materializes any J_{F_k} . It propagates the row vector $\mathbf{v}^\top := J_\ell J_{F_L} \cdots J_{F_{\ell+1}}$ right-to-left via *vector–Jacobian products* (VJPs)—one per layer, each at $O(\text{forward cost})$. Reverse-mode autodiff is precisely Theorem 4.7 with associativity exploited; we derive the algorithm formally in Chapter 18.

Chapter 5: Linear algebra I: vector spaces, basis, linear maps

Motivation

Every transformer layer is, between its bias terms and nonlinearities, a *linear map between finite-dimensional real vector spaces*. The “hidden dimension” d is just $\dim \mathbb{R}^d$; a “weight matrix” $W \in \mathbb{R}^{m \times n}$ is the matrix representation of a linear map $\mathbb{R}^n \rightarrow \mathbb{R}^m$; “residual connections” are addition in a vector space; “low-rank adapters” are statements about the image of a linear map. Before attention (Chapter 21) or embeddings (Chapter 19), we need the grammar of vector spaces, bases, dimension, kernels, images, and the rank–nullity theorem. We use Chapter 1 (sets, functions, logic) throughout.

Definitions

Definition 0.36 (Field). A field $\mathbb{F}$ is a set with operations $+, \cdot$ such that $(\mathbb{F}, +)$ is an abelian group with identity 0, $(\mathbb{F} \setminus \{0\}, \cdot)$ is an abelian group with identity 1, and multiplication distributes over addition. The canonical example is $\mathbb{F} = \mathbb{R}$.

Definition 0.37 (Vector space, eight axioms). A vector space V over $\mathbb{F}$ is a set with addition $+: V \times V \rightarrow V$ and scalar multiplication $\cdot: \mathbb{F} \times V \rightarrow V$ satisfying, for all $u, v, w \in V$ and $a, b \in \mathbb{F}$:

1. $(u + v) + w = u + (v + w)$;
2. $u + v = v + u$;
3. $\exists 0 \in V$ with $v + 0 = v$ for all v ;
4. $\forall v \exists (-v)$ with $v + (-v) = 0$;
5. $a \cdot (u + v) = a \cdot u + a \cdot v$;
6. $(a + b) \cdot v = a \cdot v + b \cdot v$;

7. $(ab) \cdot v = a \cdot (b \cdot v);$

8. $1 \cdot v = v.$

Definition 0.38 (Subspace, span, independence, basis, dimension). A subspace $U \subset V$ is a subset closed under $+$ and scalar multiplication that contains 0. A linear combination of $v_1, \dots, v_k$ is $\sum a_i v_i$ with $a_i \in \mathbb{F}$. The span of $S \subset V$ is $\text{span}(S) := \{\sum a_i v_i : v_i \in S, a_i \in \mathbb{F}\}$, the smallest subspace containing S . A finite set $\{v_1, \dots, v_k\}$ is linearly independent iff $\sum a_i v_i = 0 \Rightarrow a_1 = \dots = a_k = 0$. A basis of V is a linearly independent spanning set; the dimension $\dim V$ is its cardinality (well-defined by Theorem 0.42 below).

Definition 0.39 (Linear map, kernel, image, matrix representation). A map $T: V \rightarrow W$ between vector spaces over the same $\mathbb{F}$ is linear iff $T(au + bv) = aT(u) + bT(v)$. Define $\ker T := \{v \in V : Tv = 0\}$ and $\text{im } T := \{Tv : v \in V\}$; both are subspaces. Given bases (e_j) of $\mathbb{R}^n$ and (f_i) of $\mathbb{R}^m$, the matrix representation of T is $A \in \mathbb{R}^{m \times n}$ with $T(e_j) = \sum_i A_{ij} f_i$.

Theorems and proofs

Lemma 0.40 (Steinitz exchange). Let V be a vector space over $\mathbb{F}$. If $\{v_1, \dots, v_m\}$ is linearly independent and $\{w_1, \dots, w_n\}$ spans V , then $m \leq n$, and (after reindexing the w_j) we may replace m of them by $v_1, \dots, v_m$ so that the resulting set still spans V .

Proof. By induction on m . The case $m = 0$ is trivial. Assume the claim for $m - 1$: after reindexing, $\{v_1, \dots, v_{m-1}, w_m, \dots, w_n\}$ spans V . In particular $m - 1 \leq n$. Then

$$v_m = \sum_{i < m} a_i v_i + \sum_{j \geq m} b_j w_j$$

for some scalars. If all $b_j = 0$, then $v_m \in \text{span}(v_1, \dots, v_{m-1})$, contradicting linear independence of $\{v_1, \dots, v_m\}$. Hence some $b_{j_0} \neq 0$ for $j_0 \in \{m, \dots, n\}$, forcing $n \geq m$. Reindex so $j_0 = m$ and solve:

$$w_m = b_m^{-1} \left(v_m - \sum_{i < m} a_i v_i - \sum_{j > m} b_j w_j \right) \in \text{span}(v_1, \dots, v_m, w_{m+1}, \dots, w_n).$$

Therefore $\text{span}(v_1, \dots, v_m, w_{m+1}, \dots, w_n)$ contains $\{v_1, \dots, v_{m-1}, w_m, \dots, w_n\}$, which spans V , so the new set spans V . $\square$

Theorem 0.41 (Spanning sets contain bases; independent sets extend to bases). In a finite-dimensional V : (a) any finite spanning set contains a basis; (b) any linearly independent set extends to a basis.

Proof. (a) If a spanning set S is dependent, some $w \in S$ lies in $\text{span}(S \setminus \{w\})$, so $S \setminus \{w\}$ still spans. Iterate; finiteness terminates the process at an independent spanning set.

(b) Given an independent set L and a finite spanning set S , apply Lemma 0.40 to replace $|L|$ of the w 's by L , producing a spanning set containing L . Then apply (a), removing only S -side elements. $\square$

Theorem 0.42 (Invariance of dimension). Any two bases of a finite-dimensional V have the same cardinality.

Proof. Let $\mathcal{B}_1, \mathcal{B}_2$ be bases with $|\mathcal{B}_1| = m$, $|\mathcal{B}_2| = n$. Since $\mathcal{B}_1$ is independent and $\mathcal{B}_2$ spans, Lemma 0.40 gives $m \leq n$. Swap roles: $n \leq m$. Hence $m = n$. $\square$

Theorem 0.43 (Rank–nullity). *Let $T: V \rightarrow W$ be linear with $\dim V = n < \infty$. Then*

$$\dim \ker T + \dim \operatorname{im} T = n.$$

Proof. Let $(u_1, \dots, u_k)$ be a basis of $\ker T$. By Theorem 0.41(b), extend to a basis $(u_1, \dots, u_k, v_1, \dots, v_{n-k})$ of V . We claim $(Tv_1, \dots, Tv_{n-k})$ is a basis of $\operatorname{im} T$.

Spanning. Any $w \in \operatorname{im} T$ has $w = T(\sum a_i u_i + \sum b_j v_j) = \sum b_j Tv_j$ since $Tu_i = 0$.

Independence. If $\sum c_j Tv_j = 0$, then $T(\sum c_j v_j) = 0$, so $\sum c_j v_j \in \ker T = \operatorname{span}(u_i)$. Write $\sum c_j v_j = \sum d_i u_i$, i.e. $\sum c_j v_j - \sum d_i u_i = 0$. Independence of the full basis forces all $c_j = 0$.

Hence $\dim \operatorname{im} T = n - k = n - \dim \ker T$. $\square$

Code sketch

We will (a) implement `is_linearly_independent` and `dim_span` via `numpy.linalg.matrix_rank`; (b) build a 4×6 random integer matrix, compute rank and a basis of $\ker A$ via the right singular vectors of A corresponding to zero singular values, verifying $Av = 0$; (c) verify the change-of-basis formula $Av = P(P^{-1}AP)(P^{-1}v)$ for a random invertible P ; (d) confirm $\operatorname{rank}(A) + \dim \ker A = 6$.

Connection to LLMs

A transformer with hidden dimension d operates on the vector space $\mathbb{R}^d$. The query/key/value projections in attention and the up/down projections in the MLP are linear maps $\mathbb{R}^d \rightarrow \mathbb{R}^{d_k}$ or $\mathbb{R}^d \rightarrow \mathbb{R}^{d_v}$ (Chapter 21). The token embedding (Chapter 19) is a linear map $\mathbb{R}^{|\mathcal{V}|} \rightarrow \mathbb{R}^d$ applied to a one-hot input, i.e. a row lookup. The “rank” of an attention matrix and the “intrinsic dimension” of activations are claims about $\dim \operatorname{im}$. LoRA fine-tuning constrains weight updates to a low-dimensional subspace — a direct use of $\dim \operatorname{im} T \leq \min(m, n)$. Rank–nullity returns when we count free parameters and degrees of freedom.

Chapter 6: Linear algebra II: inner products, norms, eigenvalues, SVD

Motivation

Chapter 5 built vector spaces and linear maps as raw algebraic objects. To do *geometry* — length, angle, orthogonality, best approximation — we add a single piece of structure: an *inner product*. From it we will derive the Cauchy–Schwarz inequality, the triangle inequality, the spectral theorem for symmetric matrices, the singular value decomposition (SVD), and the Eckart–Young low-rank approximation theorem. These are the load-bearing tools behind PCA, attention, and LoRA fine-tuning of LLMs.

Definitions

Definition 0.44 (Inner product). *A function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ on a real vector space V is an inner product if for all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in V$ and $a, b \in \mathbb{R}$:*

1. *Symmetry:* $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$.
2. *Linearity in the first argument:* $\langle a\mathbf{x} + b\mathbf{y}, \mathbf{z} \rangle = a\langle \mathbf{x}, \mathbf{z} \rangle + b\langle \mathbf{y}, \mathbf{z} \rangle$.
3. *Positive-definiteness:* $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$ with equality iff $\mathbf{x} = \mathbf{0}$.

The canonical example on $\mathbb{R}^n$ is the dot product $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_i x_i y_i = \mathbf{x}^\top \mathbf{y}$.

Definition 0.45 (Induced norm). $\|\mathbf{x}\| := \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}$.

Definition 0.46 (Orthogonality). $\mathbf{x}, \mathbf{y}$ are orthogonal if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. A set $\{\mathbf{q}_i\}$ is orthonormal if $\langle \mathbf{q}_i, \mathbf{q}_j \rangle = \delta_{ij}$. An orthonormal basis is an orthonormal spanning set.

Definition 0.47 (Eigenvalue, eigenvector, characteristic polynomial). For $A \in \mathbb{R}^{n \times n}$, $(\lambda, \mathbf{v})$ with $\mathbf{v} \neq \mathbf{0}$ is an eigenpair if $A\mathbf{v} = \lambda\mathbf{v}$. The characteristic polynomial is $p_A(\lambda) := \det(A - \lambda I)$; its roots are the eigenvalues.

Definition 0.48 (Symmetric, positive (semi)definite). A is symmetric if $A^\top = A$; positive semidefinite (PSD) if symmetric with $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x}$; positive definite if strict for $\mathbf{x} \neq \mathbf{0}$.

Definition 0.49 (Singular value decomposition). A singular value decomposition of $A \in \mathbb{R}^{m \times n}$ is a factorization $A = U\Sigma V^\top$ where $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are orthogonal ($U^\top U = I$, $V^\top V = I$) and $\Sigma \in \mathbb{R}^{m \times n}$ is “diagonal” with nonnegative entries $\sigma_1 \geq \sigma_2 \geq \dots \geq 0$.

Theorems and Proofs

Theorem 0.50 (Cauchy–Schwarz). For all $\mathbf{x}, \mathbf{y} \in V$, $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$.

Proof. If $\mathbf{y} = \mathbf{0}$, both sides vanish. Otherwise, for every $t \in \mathbb{R}$,

$$0 \leq \|\mathbf{x} - t\mathbf{y}\|^2 = \langle \mathbf{x} - t\mathbf{y}, \mathbf{x} - t\mathbf{y} \rangle = \|\mathbf{x}\|^2 - 2t\langle \mathbf{x}, \mathbf{y} \rangle + t^2\|\mathbf{y}\|^2.$$

This is a quadratic in t that is nonnegative everywhere, so its discriminant satisfies

$$(2\langle \mathbf{x}, \mathbf{y} \rangle)^2 - 4\|\mathbf{x}\|^2\|\mathbf{y}\|^2 \leq 0,$$

i.e. $\langle \mathbf{x}, \mathbf{y} \rangle^2 \leq \|\mathbf{x}\|^2\|\mathbf{y}\|^2$. Taking square roots completes the proof. $\square$

Corollary 0.51 (Triangle inequality). $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$.

Proof. Expand and apply Cauchy–Schwarz:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle + \|\mathbf{y}\|^2 \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2. \quad \square$$

Theorem 0.52 (Spectral theorem, real symmetric case). Every symmetric $A \in \mathbb{R}^{n \times n}$ admits a factorization $A = Q\Lambda Q^\top$ with Q orthogonal and Λ real diagonal.

Proof. Induct on n . For $n = 1$, trivial. Assume the result for $n - 1$. The Rayleigh quotient $R(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x}$ is continuous on the unit sphere $S^{n-1} = \{\mathbf{x} : \|\mathbf{x}\| = 1\}$, which is compact, so R attains a maximum at some $\mathbf{v}_1 \in S^{n-1}$ with value λ_1 . By Lagrange multipliers (or by directly differentiating $R(\mathbf{v}_1 + t\mathbf{w})$ along any tangent $\mathbf{w} \perp \mathbf{v}_1$ at $t = 0$), $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$. The eigenvalue is real because $\lambda_1 = \mathbf{v}_1^\top A \mathbf{v}_1 \in \mathbb{R}$ and $\mathbf{v}_1 \in \mathbb{R}^n$. (Alternatively, if $A\mathbf{z} = \mu\mathbf{z}$ over $\mathbb{C}$ with $\mathbf{z} \neq \mathbf{0}$, then $\mu \bar{\mathbf{z}}^\top \mathbf{z} = \bar{\mathbf{z}}^\top A \mathbf{z} = (\overline{A\mathbf{z}})^\top \mathbf{z} = \overline{\mu} \bar{\mathbf{z}}^\top \mathbf{z}$ since A is real symmetric, so $\mu = \overline{\mu}$.)

Let $W = \{\mathbf{v}_1\}^\perp$, an $(n - 1)$ -dimensional subspace. For $\mathbf{w} \in W$,

$$\langle A\mathbf{w}, \mathbf{v}_1 \rangle = \langle \mathbf{w}, A\mathbf{v}_1 \rangle = \lambda_1 \langle \mathbf{w}, \mathbf{v}_1 \rangle = 0,$$

so A maps $W \rightarrow W$. The restriction $A|_W$ is symmetric with respect to the inherited inner product. By induction there is an orthonormal eigenbasis $\mathbf{v}_2, \dots, \mathbf{v}_n$ of W with eigenvalues $\lambda_2, \dots, \lambda_n$. Then $\mathbf{v}_1, \dots, \mathbf{v}_n$ is an orthonormal eigenbasis of A . Setting $Q = [\mathbf{v}_1 \ \dots \ \mathbf{v}_n]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ gives $A = Q\Lambda Q^\top$. $\square$

Theorem 0.53 (Existence of SVD). *Every $A \in \mathbb{R}^{m \times n}$ admits an SVD $A = U\Sigma V^\top$.*

Sketch. The matrix $A^\top A \in \mathbb{R}^{n \times n}$ is symmetric and PSD, since $\mathbf{x}^\top A^\top A \mathbf{x} = \|A\mathbf{x}\|^2 \geq 0$. By the spectral theorem $A^\top A = V\Lambda V^\top$ with V orthogonal and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$, $\lambda_i \geq 0$, sorted decreasingly. Set $\sigma_i := \sqrt{\lambda_i}$. For each i with $\sigma_i > 0$ define $\mathbf{u}_i := A\mathbf{v}_i/\sigma_i \in \mathbb{R}^m$; then

$$\langle \mathbf{u}_i, \mathbf{u}_j \rangle = \frac{\mathbf{v}_i^\top A^\top A \mathbf{v}_j}{\sigma_i \sigma_j} = \frac{\lambda_j \delta_{ij}}{\sigma_i \sigma_j} = \delta_{ij},$$

so the $\mathbf{u}_i$ are orthonormal in $\mathbb{R}^m$. Extend to an orthonormal basis to form U . By construction $AV = U\Sigma$, hence $A = U\Sigma V^\top$. $\square$

Theorem 0.54 (Eckart–Young). *Let $A = U\Sigma V^\top$ have singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. The truncated SVD $A_k := \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ minimises $\|A - B\|_F$ (and $\|A - B\|_2$) over rank- k matrices B , with $\|A - A_k\|_F^2 = \sum_{i>k} \sigma_i^2$.*

Sketch. The Frobenius norm is unitarily invariant: $\|A - B\|_F = \|U^\top(A - B)V\|_F$. The problem reduces to: among rank- k matrices $C \in \mathbb{R}^{m \times n}$, minimise

$$\|\Sigma - C\|_F^2 = \sum_i (\sigma_i - c_{ii})^2 + \sum_{i \neq j} c_{ij}^2.$$

Off-diagonal entries should be zero, and the diagonal of C should equal σ_i for $i \leq k$ and 0 for $i > k$ (matching the largest gaps), recovering A_k . The operator-norm version uses the Courant–Fischer minimax characterisation of singular values. $\square$

Code Sketch

The accompanying notebook (`cells.json`) (i) verifies Cauchy–Schwarz on 50 random pairs in $\mathbb{R}^{10}$, (ii) computes a symmetric eigendecomposition via `numpy.linalg.eigh` and checks $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ plus orthonormality, (iii) computes an SVD via `numpy.linalg.svd` and reconstructs A , and (iv) shows that Frobenius errors of rank-1 and rank-2 approximations decrease monotonically, consistent with Eckart–Young.

Connection to LLMs

Inner products and SVD are the geometric backbone of transformers.

- **Attention** (Chapters 21–22). The score matrix $QK^\top$ is the matrix of pairwise inner products of query and key embeddings. Linear-attention and Performer-style methods exploit the empirical low-rank-ness of $QK^\top$, which by Eckart–Young is best captured by truncated SVD.
- **PCA / interpretability.** SVD of an embedding matrix yields principal components; leading singular vectors expose semantic axes used in probing studies.
- **LoRA fine-tuning.** The update $W_0 + BA$ with $B \in \mathbb{R}^{m \times r}$, $A \in \mathbb{R}^{r \times n}$, $r \ll \min(m, n)$, is literally a rank- r correction; Eckart–Young guarantees its optimality at that rank in Frobenius norm.
- **Spectral norm regularisation.** Bounding $\sigma_1(W)$ controls Lipschitz constants and stabilises training.

Every later geometric statement in this book — projections, least squares, gradient flow on weight matrices, contractive maps — descends from the inequalities and decompositions proven in this chapter.

Chapter 7: Convexity and optimization; gradient descent convergence

Motivation

Chapters 3 and 4 built derivatives, the chain rule, and gradients; Chapter 6 equipped $\mathbb{R}^n$ with a norm and the Cauchy–Schwarz inequality. With those in hand we can finally ask the optimization question that drives every line of training code in this book: given $f : \mathbb{R}^n \rightarrow \mathbb{R}$, what does the iteration $x_{t+1} = x_t - \eta \nabla f(x_t)$ actually do, and when does it converge?

The honest answer for a transformer’s loss is “we don’t know”: it is non-convex, the iterates are stochastic (Chapter 13), and rigorous global convergence is open. But the *convex* case admits complete proofs, and those proofs supply the only quantitative vocabulary we have for the non-convex one — smoothness, condition number, descent, contraction. Phrases like “smooth loss landscape,” “warmup,” and “ $1/L$ step size” trace directly back to the two theorems below.

Definitions

Definition 0.55 (Convex set). $C \subseteq \mathbb{R}^n$ is convex iff for all $x, y \in C$ and $t \in [0, 1]$, $tx + (1-t)y \in C$.

Definition 0.56 (Convex function). $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex iff

$$f(tx + (1-t)y) \leq tf(x) + (1-t)f(y) \quad \forall x, y \in \mathbb{R}^n, t \in [0, 1].$$

When f is differentiable this is equivalent to the first-order characterization $f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle$: every tangent hyperplane lies below the graph.

Definition 0.57 (L -smoothness). f is L -smooth iff ∇f is L -Lipschitz: $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y , with the Euclidean norm of Chapter 6.

Definition 0.58 (μ -strong convexity). f is μ -strongly convex iff for all x, y ,

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\mu}{2}\|y - x\|^2.$$

The ratio $\kappa = L/\mu \geq 1$ is the condition number.

Definition 0.59 (Gradient descent). Fix $x_0 \in \mathbb{R}^n$, step size $\eta > 0$. The gradient descent iterates are $x_{t+1} = x_t - \eta \nabla f(x_t)$ for $t = 0, 1, 2, \dots$

Theorems and proofs

Lemma 0.60 (Descent lemma). If f is L -smooth, then for all $x, y \in \mathbb{R}^n$,

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2.$$

Proof. Let $g(t) = f(x + t(y - x))$ on $[0, 1]$. By the chain rule (Chapter 4), $g'(t) = \langle \nabla f(x + t(y - x)), y - x \rangle$, and the fundamental theorem of calculus gives

$$f(y) - f(x) = g(1) - g(0) = \int_0^1 \langle \nabla f(x + t(y - x)), y - x \rangle dt.$$

Subtract $\langle \nabla f(x), y - x \rangle$ from both sides:

$$f(y) - f(x) - \langle \nabla f(x), y - x \rangle = \int_0^1 \langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle dt.$$

By Cauchy–Schwarz (Chapter 6) and the Lipschitz hypothesis,

$$\langle \nabla f(x + t(y - x)) - \nabla f(x), y - x \rangle \leq \|\nabla f(x + t(y - x)) - \nabla f(x)\| \|y - x\| \leq Lt\|y - x\|^2.$$

Integrating, $\int_0^1 Lt\|y - x\|^2 dt = \frac{L}{2}\|y - x\|^2$. $\square$

Theorem 0.61 (GD on L -smooth convex). *Let f be convex and L -smooth with minimizer x^* and minimum $f^* = f(x^*)$. Then gradient descent with $\eta = 1/L$ satisfies*

$$f(x_T) - f^* \leq \frac{L\|x_0 - x^*\|^2}{2T}, \quad T \geq 1.$$

Proof. Apply the descent lemma at $y = x_{t+1} = x_t - \frac{1}{L}\nabla f(x_t)$, $x = x_t$:

$$f(x_{t+1}) \leq f(x_t) - \frac{1}{2L}\|\nabla f(x_t)\|^2. \quad (\star)$$

In particular $f(x_t)$ is non-increasing in t . Convexity at x_t versus x^* gives $f(x_t) - f^* \leq \langle \nabla f(x_t), x_t - x^* \rangle$. Expand the gradient step:

$$\|x_{t+1} - x^*\|^2 = \|x_t - x^*\|^2 - \frac{2}{L}\langle \nabla f(x_t), x_t - x^* \rangle + \frac{1}{L^2}\|\nabla f(x_t)\|^2,$$

so

$$\frac{2}{L}\langle \nabla f(x_t), x_t - x^* \rangle = \|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2 + \frac{1}{L^2}\|\nabla f(x_t)\|^2,$$

hence

$$f(x_t) - f^* \leq \frac{L}{2}(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2) + \frac{1}{2L}\|\nabla f(x_t)\|^2.$$

Adding $(\star)$ rewritten as $\frac{1}{2L}\|\nabla f(x_t)\|^2 \leq f(x_t) - f(x_{t+1})$ to the right-hand side and rearranging,

$$f(x_{t+1}) - f^* \leq \frac{L}{2}(\|x_t - x^*\|^2 - \|x_{t+1} - x^*\|^2).$$

Telescope $t = 0, \dots, T-1$:

$$\sum_{t=0}^{T-1} (f(x_{t+1}) - f^*) \leq \frac{L}{2}\|x_0 - x^*\|^2.$$

Since $f(x_t)$ is non-increasing, $T(f(x_T) - f^*) \leq \sum_{t=0}^{T-1} (f(x_{t+1}) - f^*)$, which yields the claim. $\square$

Theorem 0.62 (GD on L -smooth μ -strongly convex). *Let f be μ -strongly convex and L -smooth, with $\eta = 1/L$. Then*

$$\|x_t - x^*\|^2 \leq (1 - \mu/L)^t \|x_0 - x^*\|^2.$$

Proof. Strong convexity at x_t versus x^* rearranges to

$$\langle \nabla f(x_t), x_t - x^* \rangle \geq f(x_t) - f^* + \frac{\mu}{2}\|x_t - x^*\|^2,$$

and from $(\star)$ in the previous proof, $\frac{1}{2L}\|\nabla f(x_t)\|^2 \leq f(x_t) - f(x_{t+1}) \leq f(x_t) - f^*$, i.e. $\frac{1}{L^2}\|\nabla f(x_t)\|^2 \leq \frac{2}{L}(f(x_t) - f^*)$. Substituting both into the gradient-step expansion,

$$\begin{aligned} \|x_{t+1} - x^*\|^2 &= \|x_t - x^*\|^2 - \frac{2}{L}\langle \nabla f(x_t), x_t - x^* \rangle + \frac{1}{L^2}\|\nabla f(x_t)\|^2 \\ &\leq \|x_t - x^*\|^2 - \frac{2}{L}(f(x_t) - f^* + \frac{\mu}{2}\|x_t - x^*\|^2) + \frac{2}{L}(f(x_t) - f^*) \\ &= (1 - \frac{\mu}{L})\|x_t - x^*\|^2. \end{aligned}$$

Iterate. $\square$

Remark 0.63. The contraction factor $1 - 1/\kappa$ degrades as the condition number $\kappa = L/\mu$ grows: ill-conditioned problems converge slowly. The optimal step $\eta = 2/(L + \mu)$ improves the per-step factor to $((\kappa - 1)/(\kappa + 1))^2$, but does not change the $O(\kappa \log(1/\varepsilon))$ iteration complexity scaling.

Code sketch

Instantiate $f(x, y) = (x - 1)^2 + 2(y + 1)^2$, a strongly convex quadratic with Hessian $\text{diag}(2, 4)$, so $\mu = 2$, $L = 4$. Gradient descent with $\eta = 1/L$ contracts $\|x_t - x^*\|^2$ by factor $1 - \mu/L = 1/2$ per step. For a degenerate $f(x) = \|Ax - b\|^2$ with $A \in \mathbb{R}^{20 \times 10}$ ill-conditioned, the strong-convexity rate degrades to the convex $O(1/T)$, visible as slope ≈ -1 on a log-log plot of $f(x_T) - f^*$ versus T . A side-by-side run with $\eta = 1/L$ and the optimal $\eta = 2/(L + \mu)$ confirms the sharper contraction predicted by the remark.

Connection to LLMs

Transformer losses are spectacularly non-convex: permuting attention heads or flipping signs in a layernorm gives an equivalent minimum. None of the theorems above apply globally. They apply *locally*, and they supply the operating intuitions that survive the jump.

- **Smooth loss landscape.** The descent lemma says $\eta < 2/L$ is sufficient for monotone decrease. Empirically estimating local L via the Hessian's top eigenvalue is exactly what learning-rate range tests and gradient-norm clipping approximate.
- **Warmup and LR scheduling** (Chapter 27). Early in training the local L is large (sharp minima nearby); a small η avoids the quadratic blow-up term in the descent lemma. As the trajectory enters flatter regions, L drops and η can grow.
- **SGD and AdamW** (Chapters 13, 14). Stochastic gradients break the deterministic descent lemma; the convergence proofs replace $(\star)$ with an expectation inequality plus a variance term, but the algebraic skeleton — descent lemma, telescoping, contraction — is preserved.

The convex theory is a load-bearing analogy: the only place where the rates are honest, and the conceptual scaffold for every heuristic layered on top.

Block B — Probability and Information

Chapter 8: Probability foundations: sample spaces, sigma-algebras, Kolmogorov axioms

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 9: Random variables, distributions, CDF/PMF/PDF

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 10: Expectation, variance, covariance; Jensen's inequality

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 11: Information theory: self-information, entropy, cross-entropy, KL

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 12: Statistical inference: likelihood, MLE, ERM, bias-variance

Stub chapter — will be filled by Phase 2 chapter agent.

Block C — Stochastic Optimization

Chapter 13: SGD: stochastic-approximation theorem; mini-batching; convergence sketch

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 14: Momentum, RMSProp, AdamW: derivation and bias-correction proof

Stub chapter — will be filled by Phase 2 chapter agent.

Block D — Neural Networks

Chapter 15: MLPs as compositional functions; universal approximation

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 16: Activation functions: ReLU/GELU/softmax with derivatives

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 17: Loss functions: MSE, cross-entropy; gradients from first principles

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 18: Backpropagation: chain rule applied; reverse-mode AD as a graph algorithm

Stub chapter — will be filled by Phase 2 chapter agent.

Block E — Sequence Models and Attention

Chapter 19: Embeddings: token to vector; lookup as a linear map; weight tying

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 20: RNN intuition; vanishing-gradient proof; why we need attention

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 21: Scaled dot-product attention: derivation, softmax-temperature analysis

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 22: Multi-head attention: parallel heads as concat-then-project; complexity

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 23: Transformer block: residual + LayerNorm/RMSNorm + FFN + attention; gradient-flow argument

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 24: Positional encoding: sinusoidal derivation, RoPE construction

Stub chapter — will be filled by Phase 2 chapter agent.

Block F — Pre-training

Chapter 25: Causal masking; next-token prediction loss as MLE on the empirical distribution

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 26: Tokenization: BPE algorithm; greedy merge correctness

Stub chapter — will be filled by Phase 2 chapter agent.

Chapter 27: Pre-training pipeline: AdamW + warmup + cosine decay + gradient clipping; tiny-GPT training run

Stub chapter — will be filled by Phase 2 chapter agent.

Block G — Post-training

Chapter 28: SFT, RLHF (PPO/GRPO), and DPO; train + post-train a tiny GPT

Stub chapter — will be filled by Phase 2 chapter agent.